

BUSINESS REPORT – MACHINE LEARNING-1 **PROJECT**

Casestudy - **INNHOTELS GROUP**

Submitted by: Goli Bhagyasri

PROBLEM STATEMENT:

Context:

A significant number of hotel bookings are called off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost, which is beneficial to hotel guests, but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations.

The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behavior. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impacts a hotel on various fronts:

- Loss of resources (revenue) when the hotel cannot resell the room.
- Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
- Lowering prices last minute, so the hotel can resell a room, resulting in a reduction of the profit margin.
- Human resources to make arrangements for the guests.

Objective:

The increasing number of cancellations calls for a Machine Learning based solution that can help in predicting which booking is likely to be canceled. INN Hotels Group has a chain of hotels in Portugal, and they are facing problems with the high number of booking cancellations and have reached out to your firm for data-driven solutions. As a data scientist have to analyze the data provided to find which factors have a high influence on booking cancellations, build a predictive model that can predict which bookings are going to be canceled in advance, and help in formulating profitable policies for cancellations and refunds.

Data Description:

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

Data Dictionary:

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel

- `type_of_meal_plan`: Type of meal plan booked by the customer:
 - Not Selected – No meal plan selected
 - Meal Plan 1 – Breakfast
 - Meal Plan 2 – Half board (breakfast and one other meal)
 - Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- `required_car_parking_space`: Does the customer require a car parking space? (0 - No, 1- Yes)
- `room_type_reserved`: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- `lead_time`: Number of days between the date of booking and the arrival date
- `arrival_year`: Year of arrival date
- `arrival_month`: Month of arrival date
- `arrival_date`: Date of the month
- `market_segment_type`: Market segment designation.
- `repeated_guest`: Is the customer a repeated guest? (0 - No, 1- Yes)
- `no_of_previous_cancellations`: Number of previous bookings that were canceled by the customer prior to the current booking
- `no_of_previous_bookings_not_canceled`: Number of previous bookings not canceled by the customer prior to the current booking
- `avg_price_per_room`: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- `no_of_special_requests`: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- `booking_status`: Flag indicating if the booking was canceled or not.

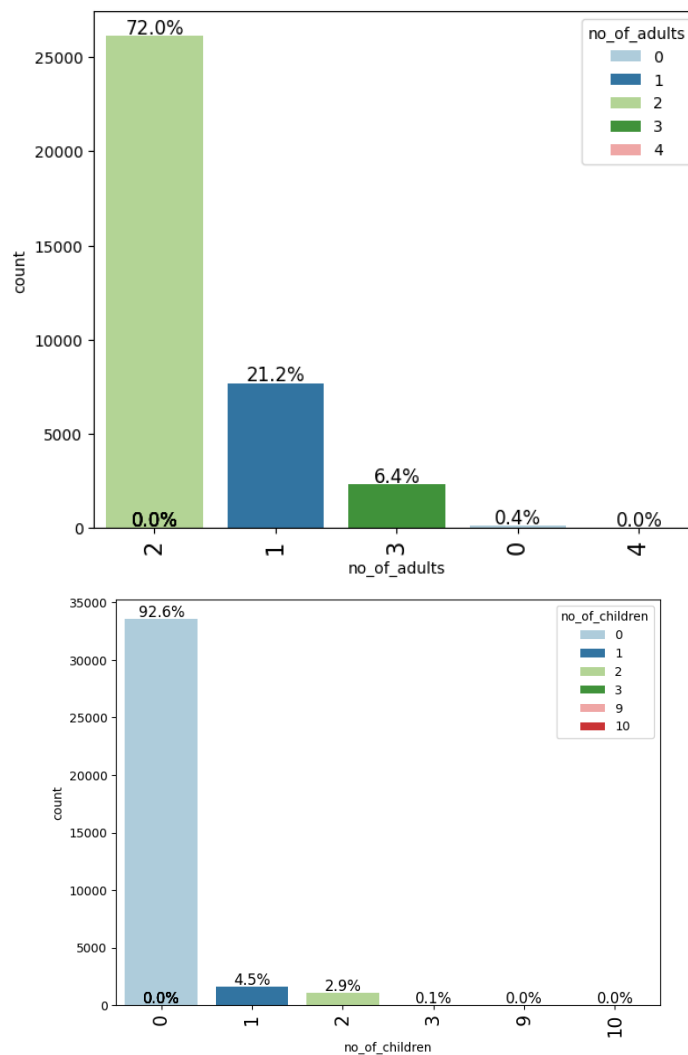
Exploratory Data Analysis:

Problem statement:

To build a predictive model that can predict which bookings are going to be cancelled in advance, and help in formulating profitable policies for cancellations and refunds.

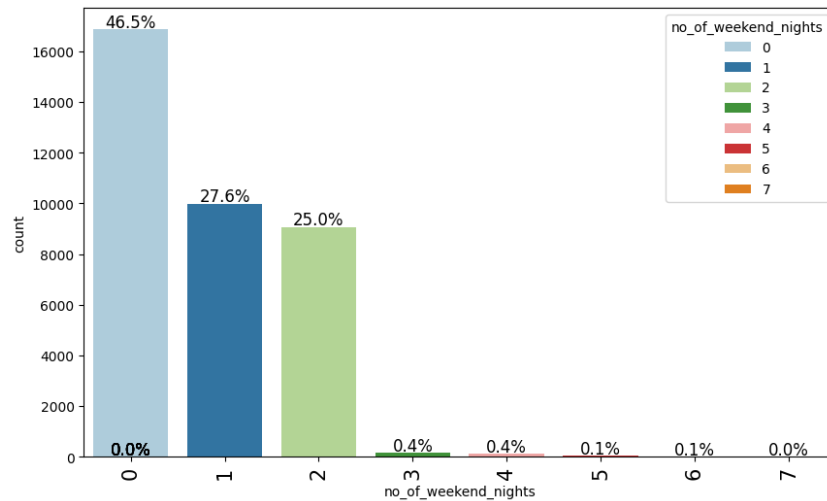
Univariate Analysis:

1. Observations on no_of_adults and no_of_children

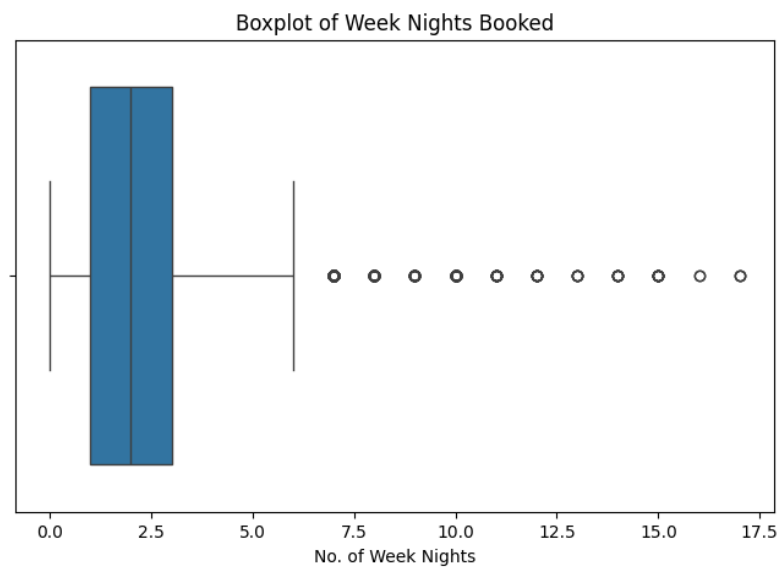


- It can be seen that no_of_adults visiting as 2 people is 72%.
- Around majority ie., 93% of visitors have no children which can be later checked for correlation with other variables and if no significance is seen, this variable can be dropped.

2. Observations on no_of_weekend_nights and no_of_week_nights

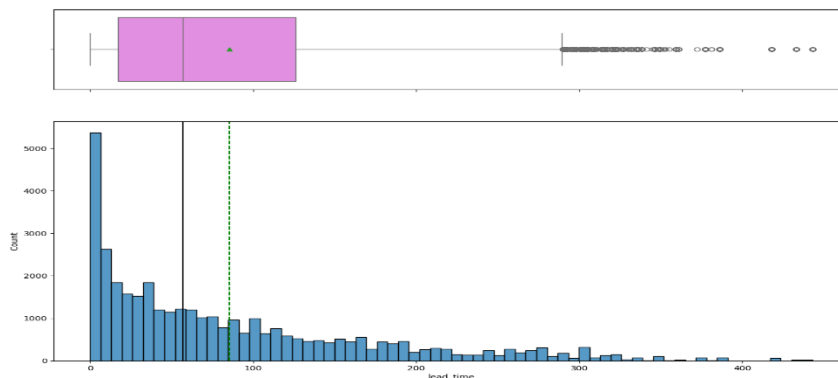


- It is evident that around 52% visitors stay for 1 or 2 weekend days



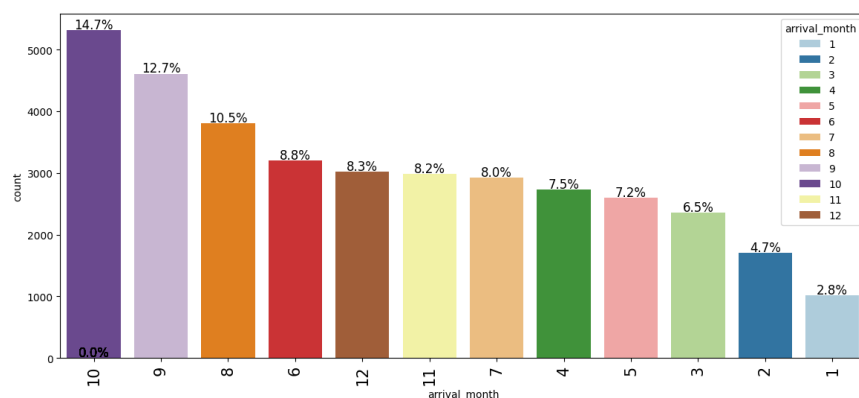
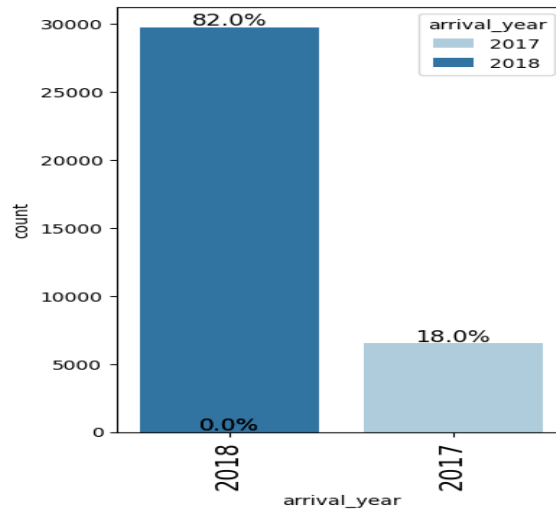
- Graph shows a lot of data points towards the extreme right, showing the presence of outliers. It is suggested to treat the outliers as the data is heavily right skewed.

3. Observations on lead_time



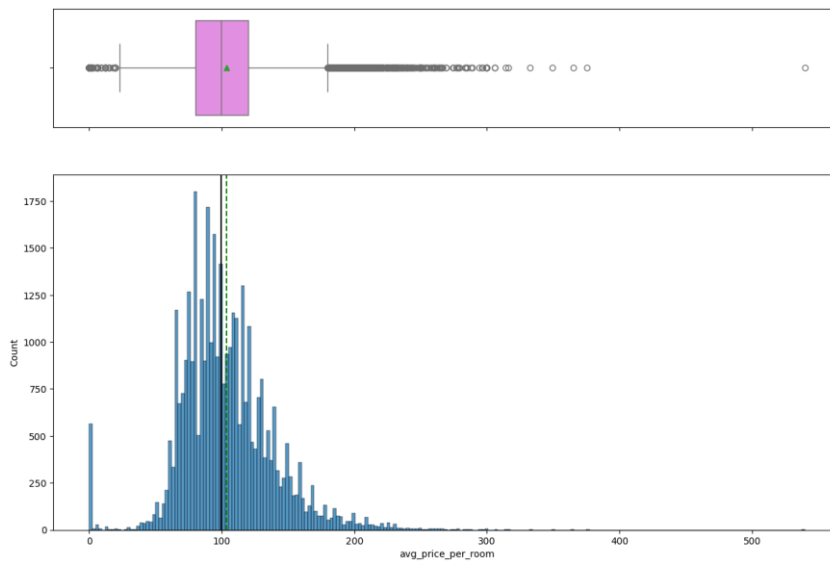
- Graph shows a lot of data points towards the extreme right, showing the presence of outliers. It is suggested to treat the outliers as the data is heavily right skewed.

4. Observations on arrival_year, arrival_month



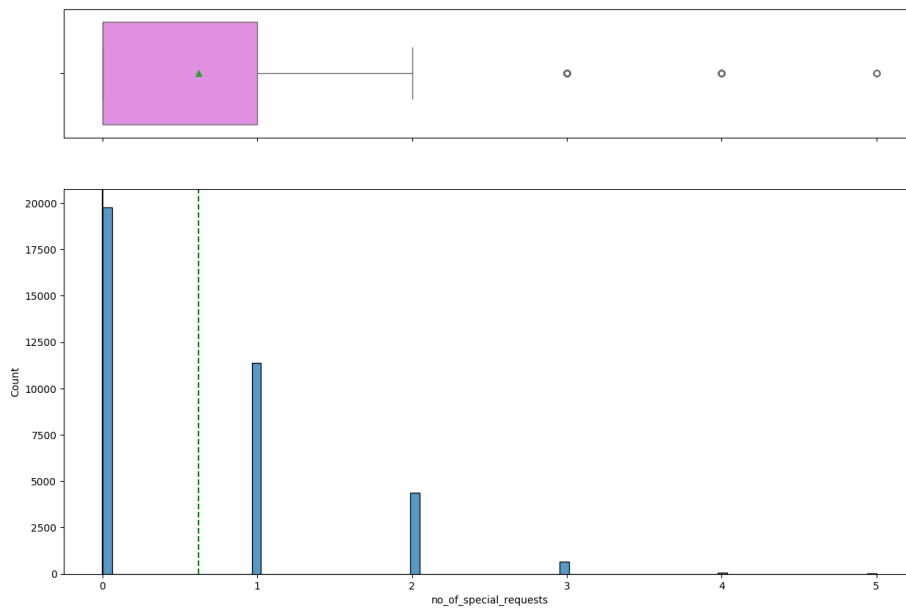
- 82% of the visitors visited in 2018 amongst which most of them visited during September and October.
- Hence busiest months in the hotel are October, september and August.

5. Observations on avg_price_per_room



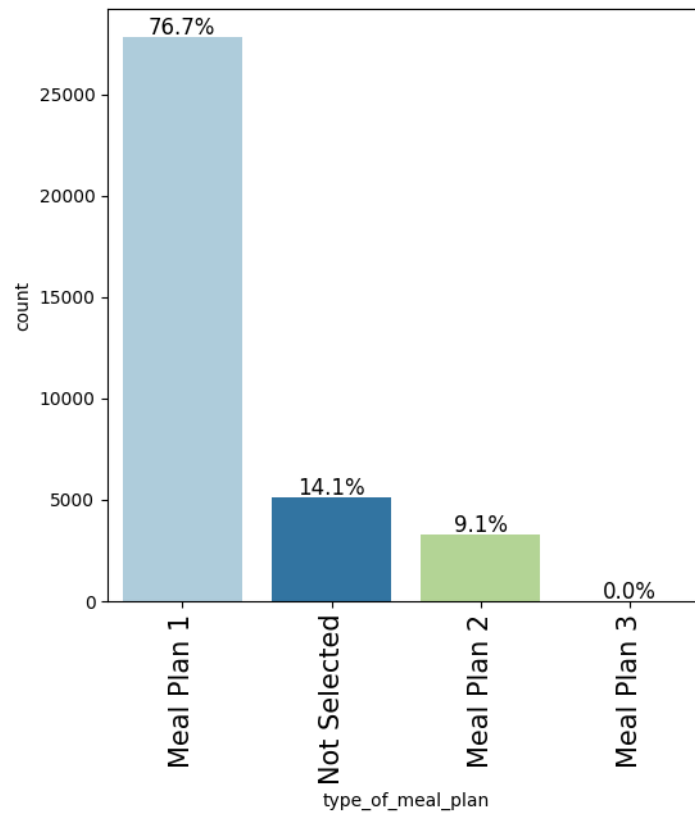
- Data shows extreme outliers at both the end and cab be treated.

6. Observations on no_of_special_requests

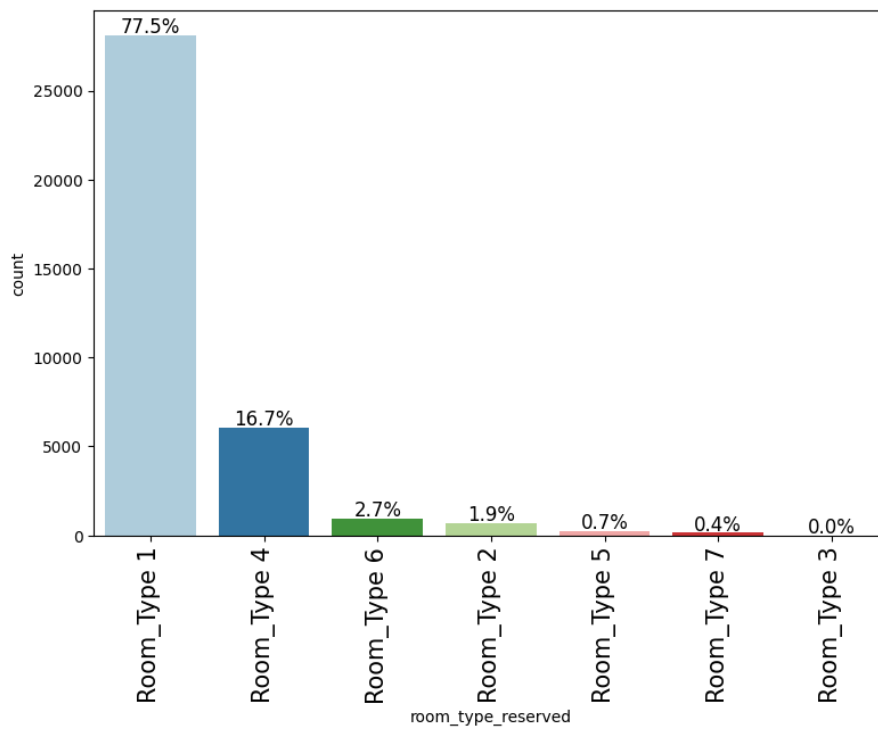


- Data shows extreme outliers at both the end and cab be treated.

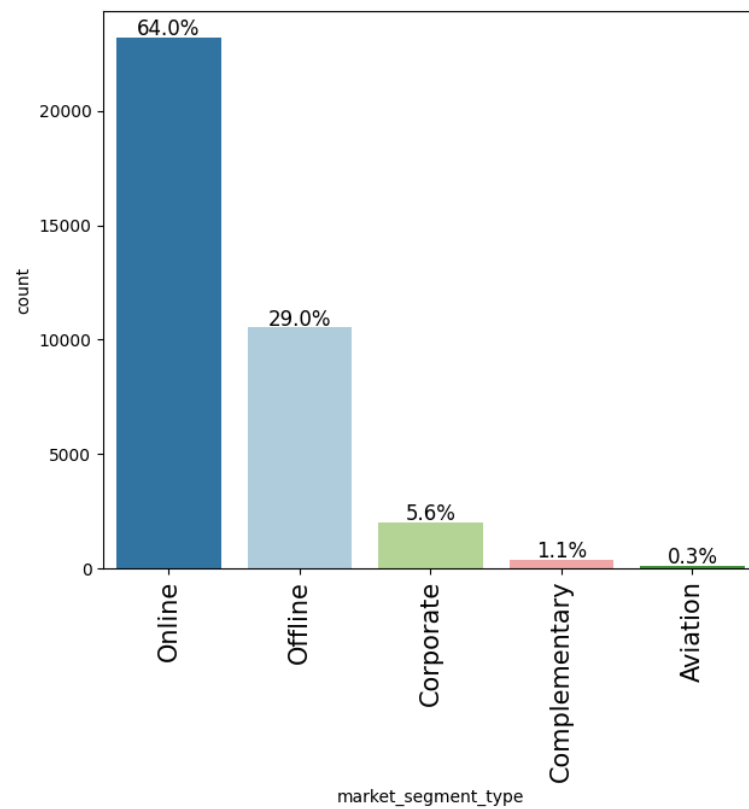
7. Observations on type_of_meal_plan



8. Observations on room_type_reserved

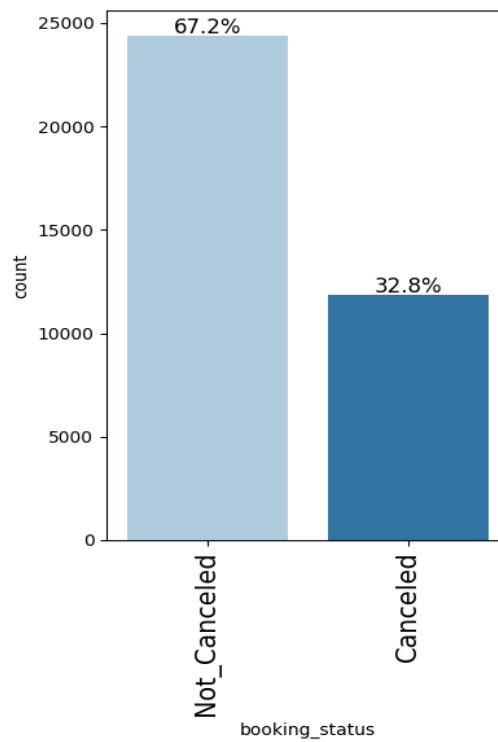


9. Observations on market_segment_type



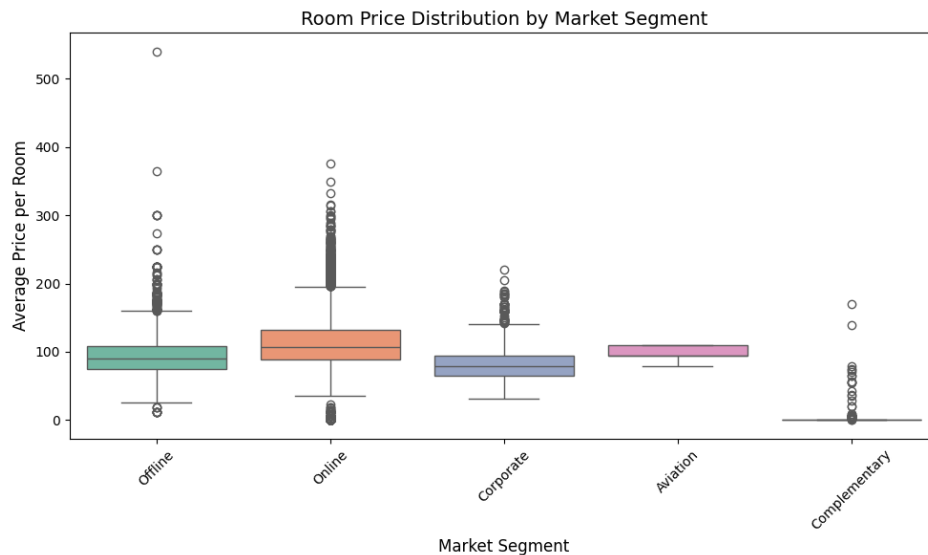
- Most of the guests come from online market segment

10. Observations on booking_status



- 32% of the people who booked have cancelled their rooms
- Hence it is better to study the correlation of this variable with other variables to understand what other variables affect this variable

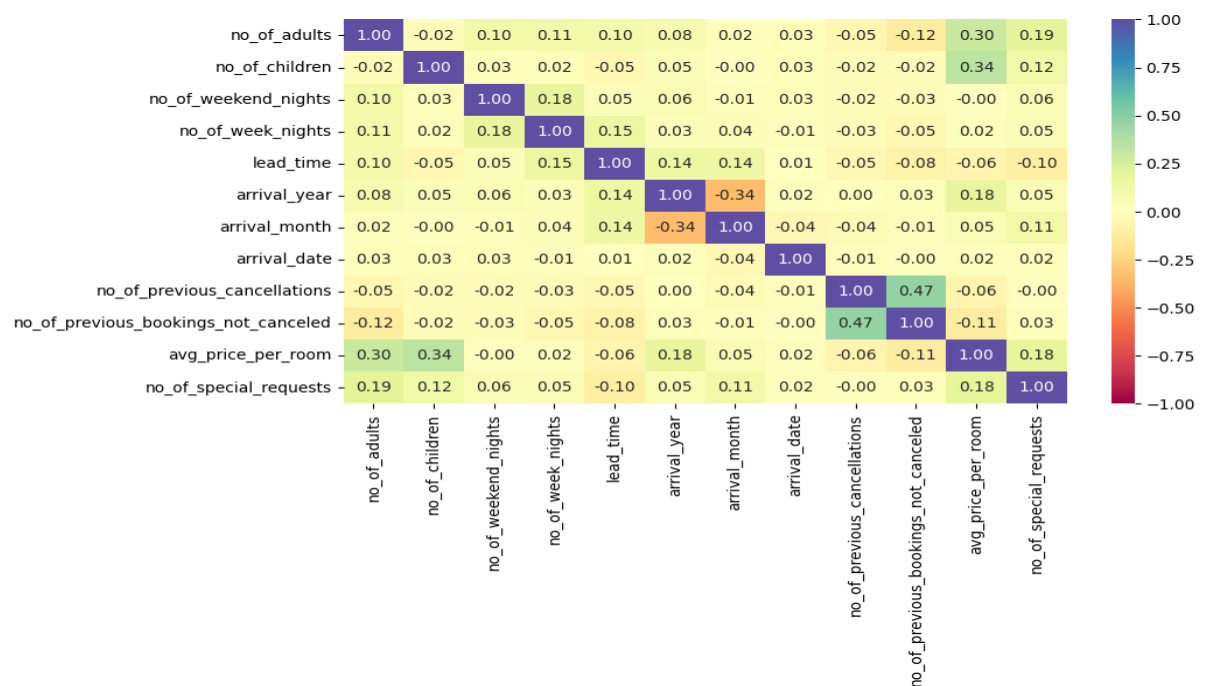
11. Observations on room prices across segments



- Rooms booked online and offline have higher prices compared to corporate and other market segments

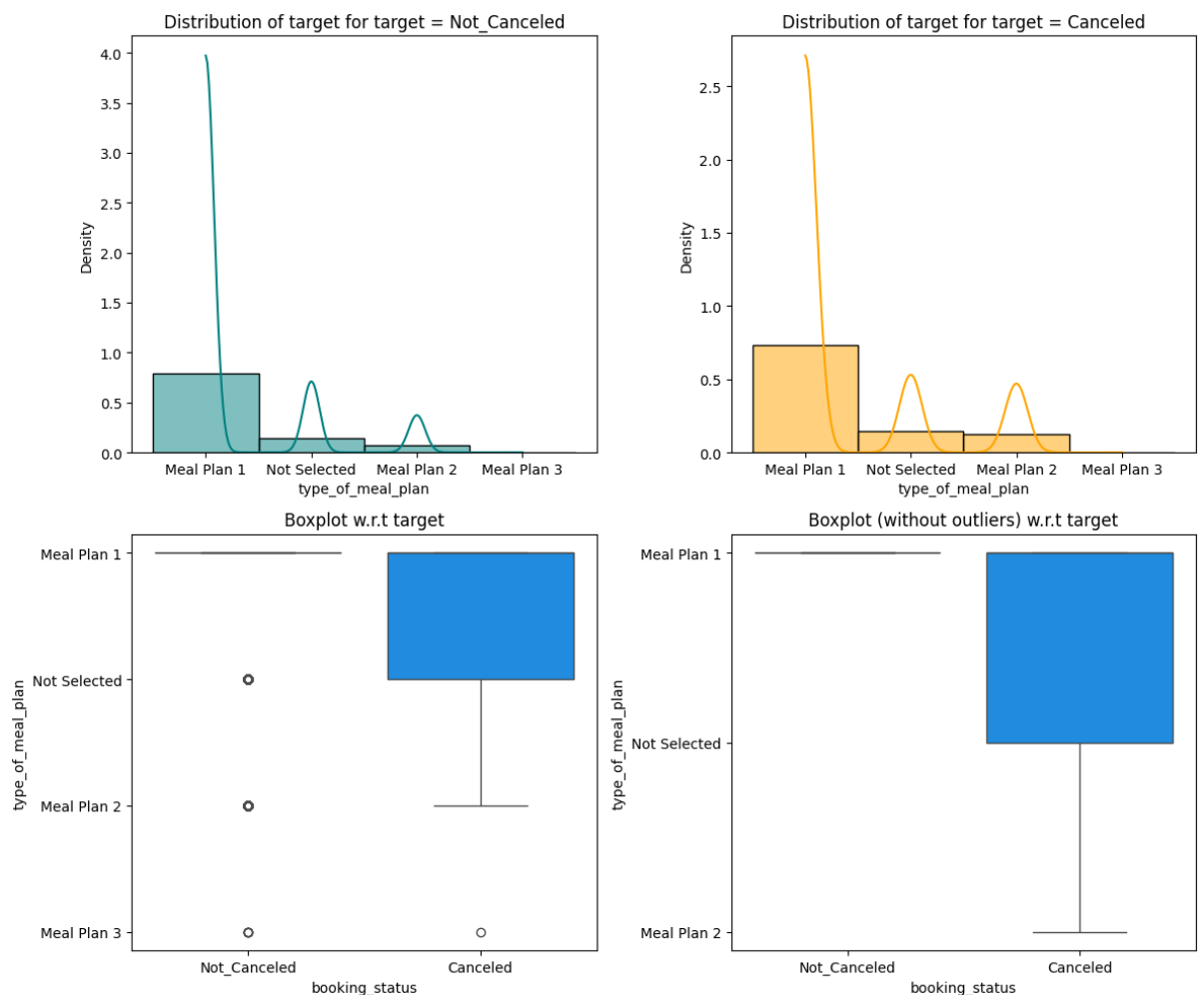
Bivariate Analysis:

1. Correlation Analysis



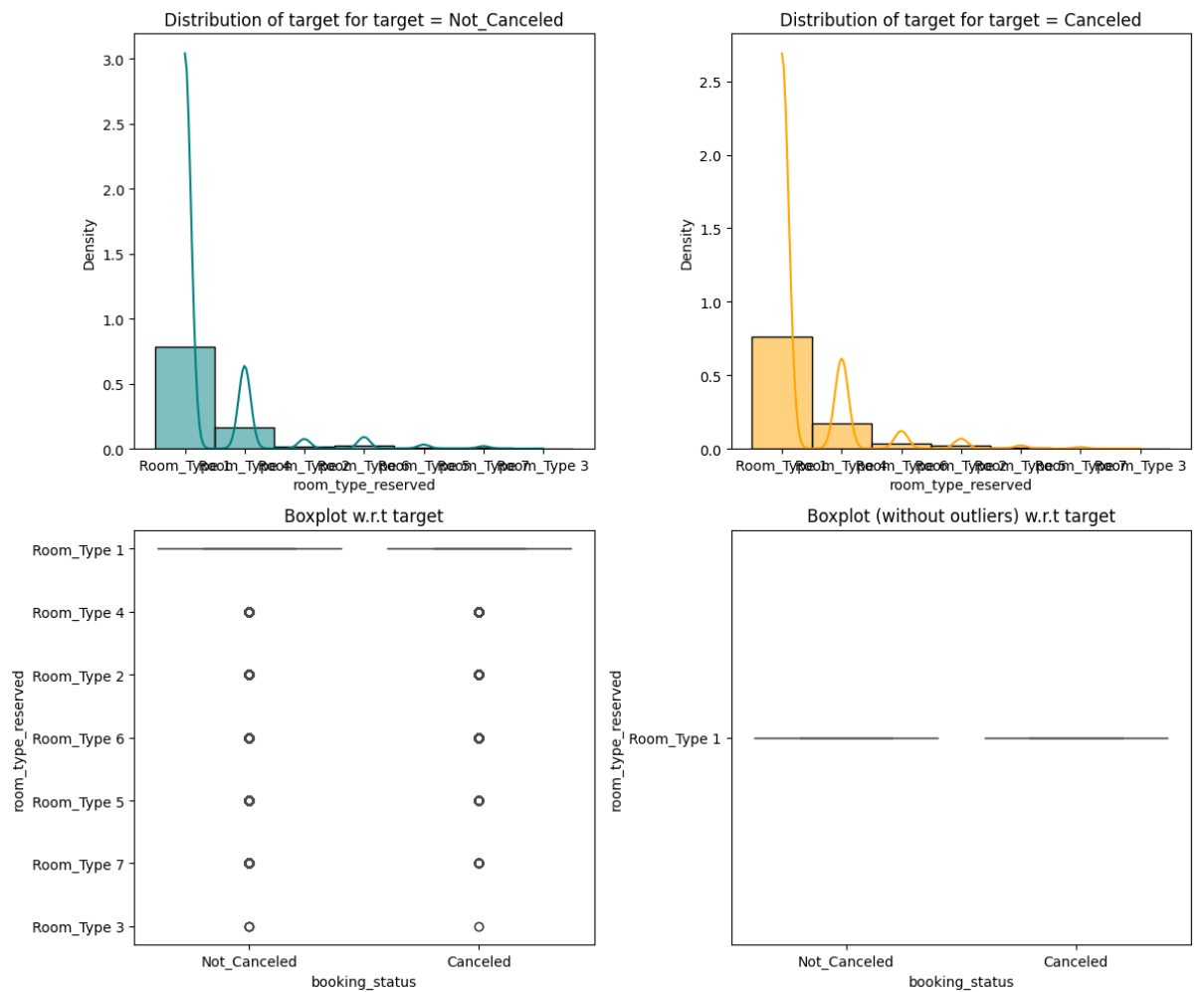
- no_of_adults vs avg_price_per_room indicates more adults often mean bigger or pricier room types
- lead_time is not strongly correlated with most variables—this could be an independent predictor of cancellations.
- no_of_previous_cancellations and no_of_previous_bookings_not_canceled have a moderate positive correlation 0.47, meaning customers with more booking history show both types of behavior.

2. Observations on booking_status vs type_of_meal_plan



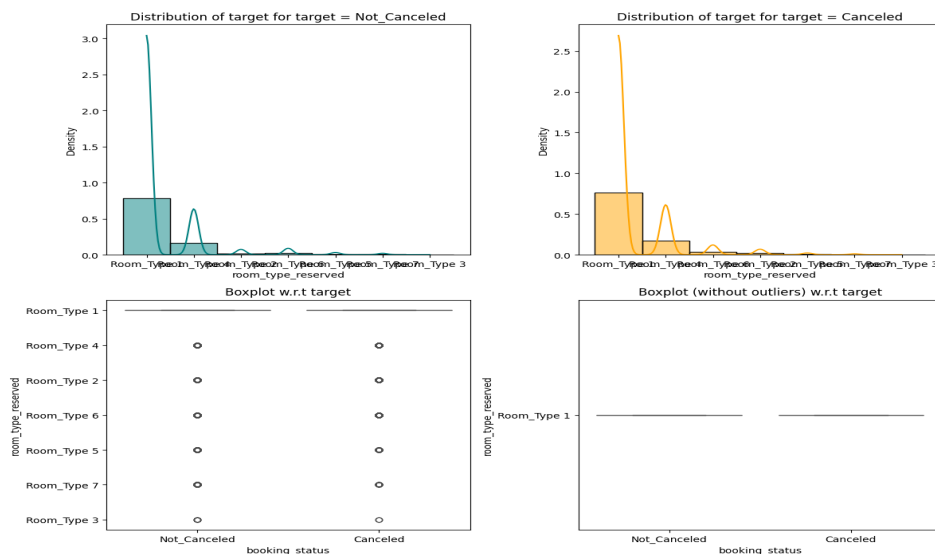
- There is no relationship between booking_status and type_of_meal_plan

3. Observations on booking_status vs room_type_reserved



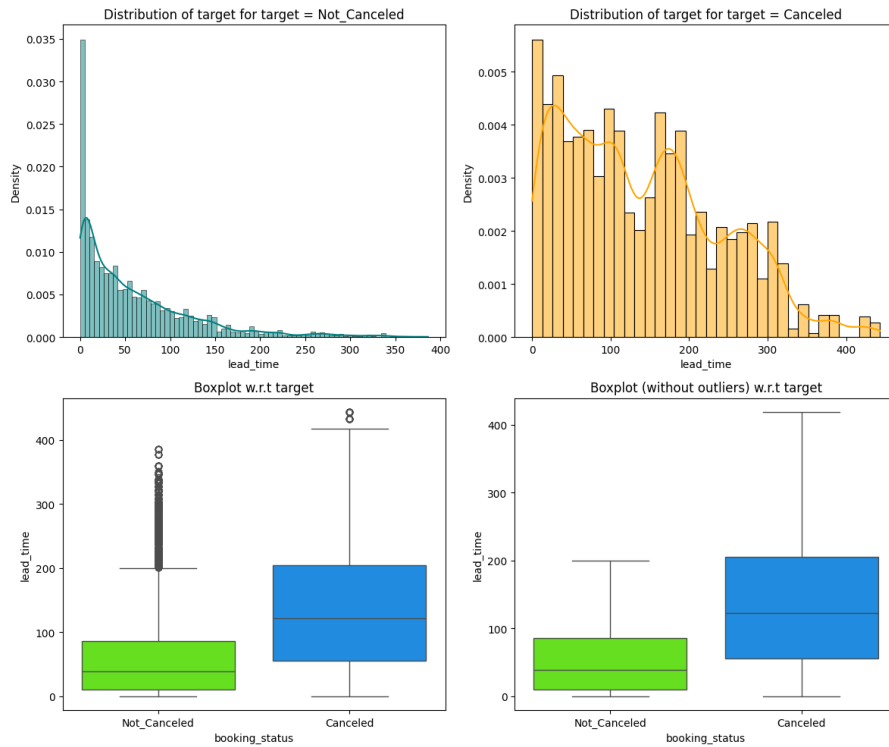
- There is no relationship between booking_status and room type

4. Observations on booking_Status vs market_segment_type



- There is no relationship between booking_status and market segment type

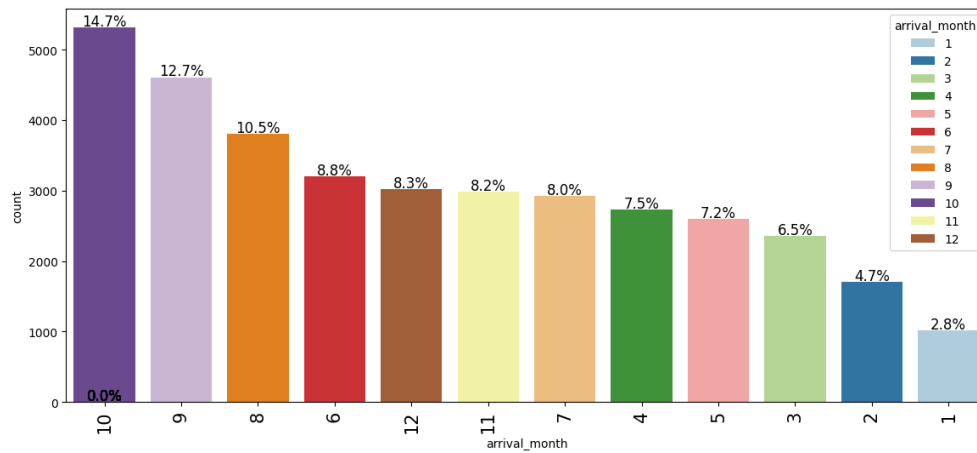
5. Observations on booking_Status vs lead_time



- There is a clear relationship between leadtime and booking status. There are more cancellations when leadtime is longer.

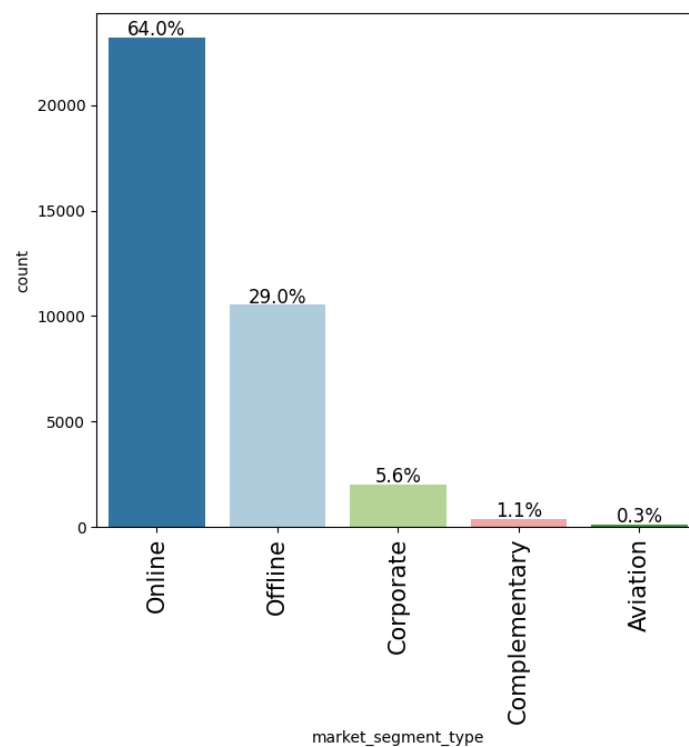
EDA Questions:

1. What are the busiest months in the hotel?



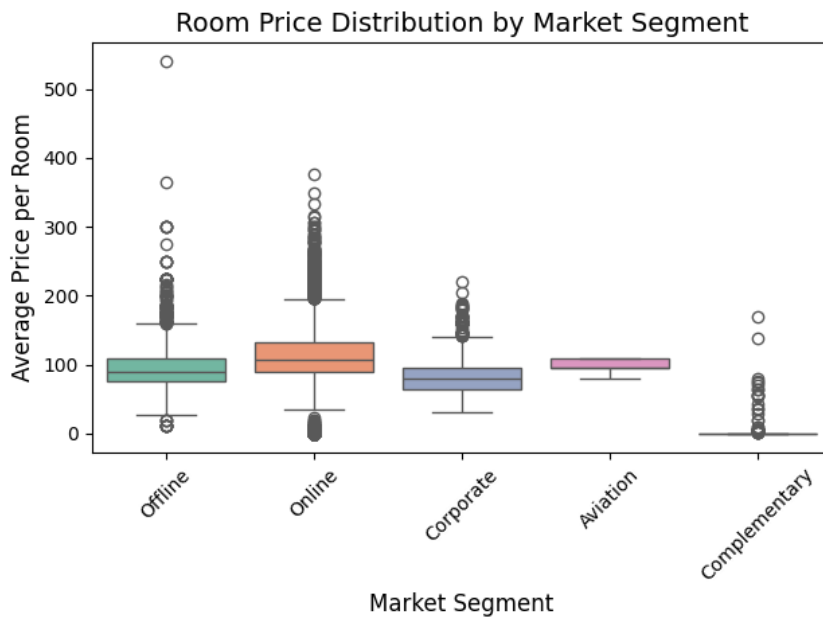
Ans: Hence busiest months in the hotel are October, September and August.

2. Which market segment do most of the guests come from?



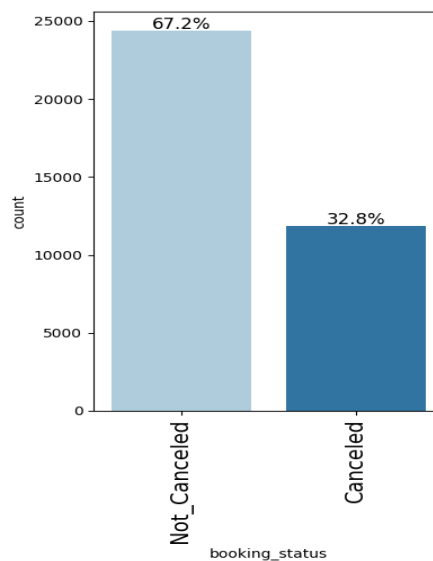
Ans: Most of the guests come from online market segment as per the above graph.

3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?



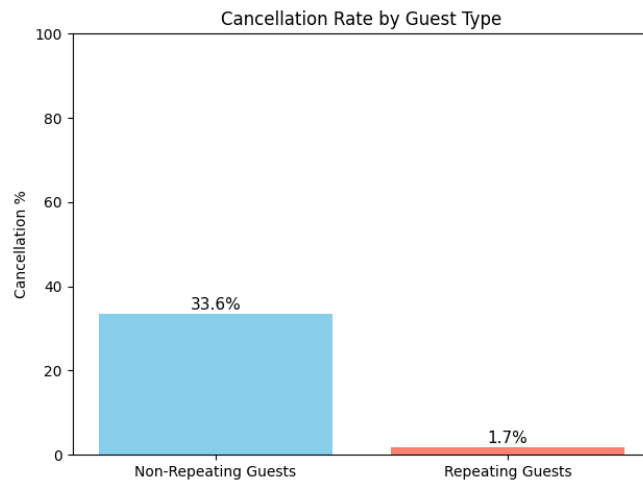
Ans: Rooms booked online and offline have higher prices compared to corporate and other market segments.

4. What percentage of bookings are canceled?



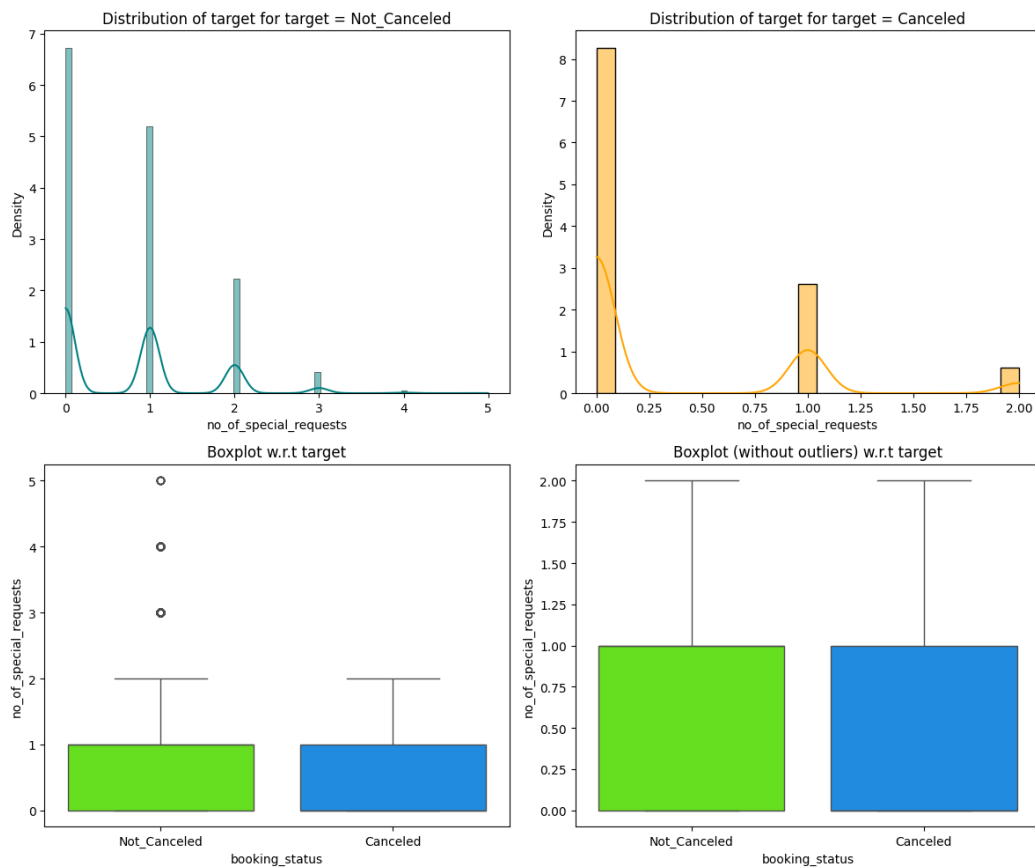
Ans: 32.8% of bookings got cancelled.

5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?



Ans: Only around 1.7% of repeating guests cancel the bookings.

6. Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

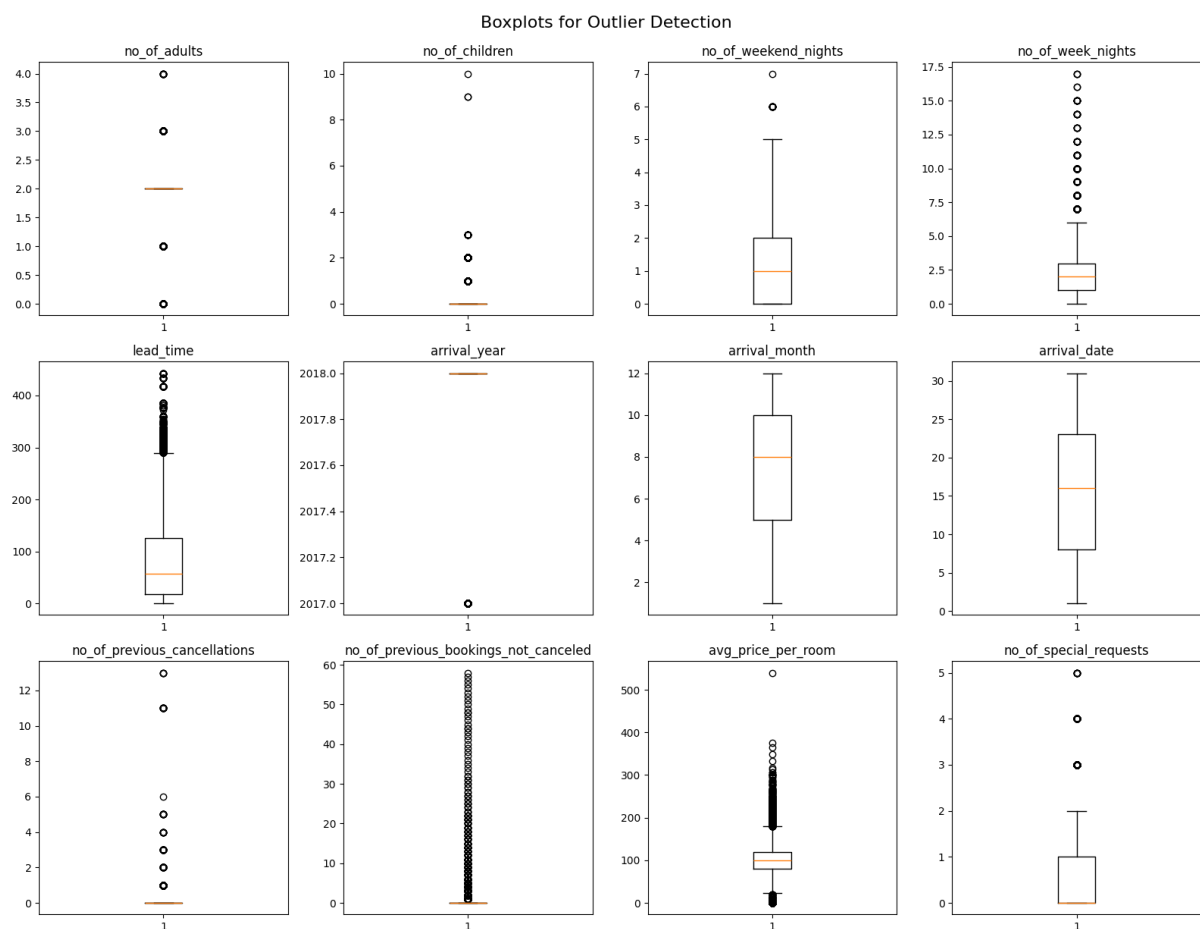


Ans: No of special requests don't significantly affect the booking cancellation.

DATA PREPARATION:

- We want to predict which feature is more likely to affect the booking cancellation.
- Before we proceed to build a model, we'll have to encode categorical features.
- We'll split the data into train and test to be able to evaluate the model that we build on the train data.

1.Outliers detection and treatment:



- There are quite a few outliers in the data.
- we will not treat them in the next step as they are proper values

2. Defined features and Onehot-Encoding:

- we did onehot-encoding for "type_of_meal_plan", "room_type_reserved", "market_segment_type" features.
- And split the data to training and testing data, added the intercepts to check and perform the model building.

Results:

Shape of Training set : (25392, 25)

Shape of test set : (10883, 25)

Percentage of classes in training set:

booking_status

0 0.672692

1 0.327308

Percentage of classes in test set:

booking_status

0 0.671598

1 0.328402

Logistic Regression (with Statsmodel):

Logistic regression is a **classification algorithm** that models the **probability** that a given input belongs to a particular category (e.g., cancel vs. not cancel). Unlike linear regression, it outputs values between 0 and 1 using the **logit (sigmoid)** function.

- We scale the data
- Add the constant and check the logistics regression using logitcreg formula

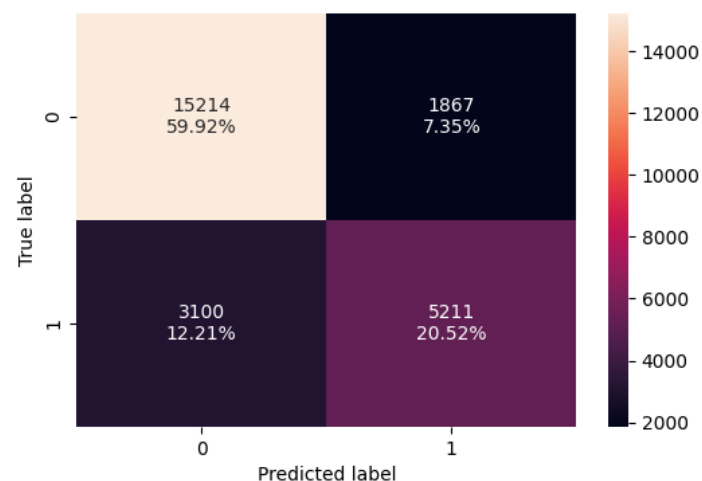
Inferences:

- Positive & significant influence indicates likely strong predictors that increase the odds of cancellation.
- Mixed influence, all significant influence says some positively, some negatively affect outcome.
- Strong negative coefficients, highly significant says likely dummies or encoded categories that decrease cancellation odds drastically

Training performance:

	Accuracy	Recall	Precision	F1
0	0.804387	0.627	0.736225	0.677237

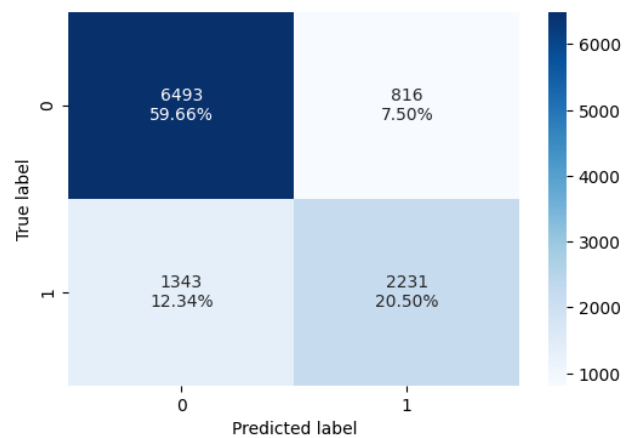
Confusion matrix:



Testing set performance:

	Accuracy	Recall	Precision	F1
0	0.801617	0.624231	0.732196	0.673916

Confusion matrix:



The Logistic Regression model has a average performance on both training and test sets.

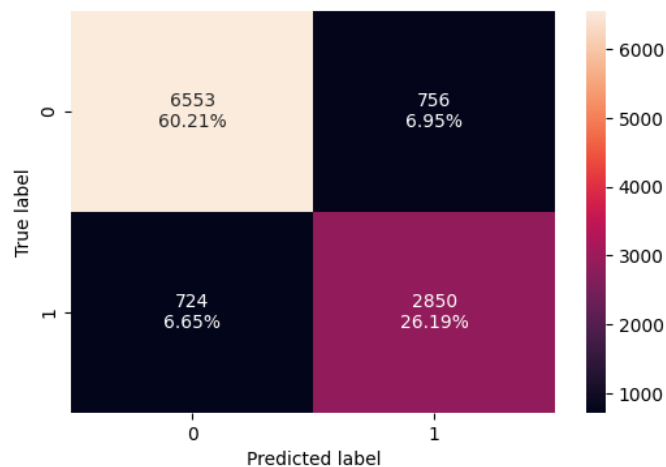
Decision tree classifier:

A decision tree is a flowchart-like structure used for classification or regression. It splits the data into branches based on feature values, making decisions at each node, until it lands at a final prediction (leaf node)

Test data :

	Accuracy	Recall	Precision	F1
0	0.864008	0.797426	0.790349	0.793872

Confusion matrix:



Observations

- The f1_score of the model is ~0.79 and we will try to maximize it further
- The variables used to build the model might contain multicollinearity, which will affect the p-values
- We will have to remove multicollinearity from the data to get reliable coefficients and p-values
- The decision tree model is overfitting the data as expected and not able to generalize well on the test set.
- We will have to prune the decision tree.

Model can make wrong predictions as:

Predicting a booking cancellation as not cancelled and not cancelled as cancellations

Which case is more important?

- If we predict a guest does cancel, but the model predicts they will not. we might hold a room unnecessarily, turn away other potential guests, lose revenue due to last-minute vacancy. We may also lost opportunity and unpreparedness. This is called False negatives.
- If we predict a guest did not cancel as cancelled, we can offer the room to others, overbook to compensate. This is called False positive.

**predicting a cancellation as not canceled (false negative) is worse, because it can as it leads to unsold inventory, reduces revenue and disruptive planning. **

How to reduce this loss?

- We need to reduce False Negatives
- f1_score should be maximized as the greater the f1_score, the higher the chances of reducing False Negatives
 - fi_score is computed as

$$f1_score = \frac{2 * Precision * Recall}{Precision + Recall}$$

“Logistic Regression (deal with multicollinearity, remove high p-value variables, determine optimal threshold using ROC curve)”

Detecting and Dealing with Multicollinearity:

Data Preparation And Data modeling

- We want to predict which feature is more likely to affect the booking cancellation.
- Before we proceed to build a model, we'll have to encode categorical features.
- We'll split the data into train and test to be able to evaluate the model that we build on the train data.

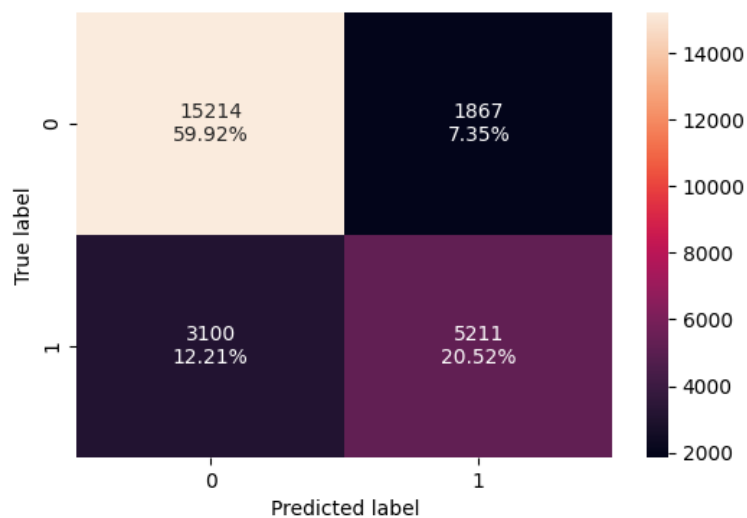
Inferences:

- Positive & significant influence indicates likely strong predictors that increase the odds of cancellation.

- Mixed influence, all significant influence says some positively, some negatively affect outcome.
- Strong negative coefficients, highly significant says likely dummies or encoded categories that decrease cancellation odds drastically

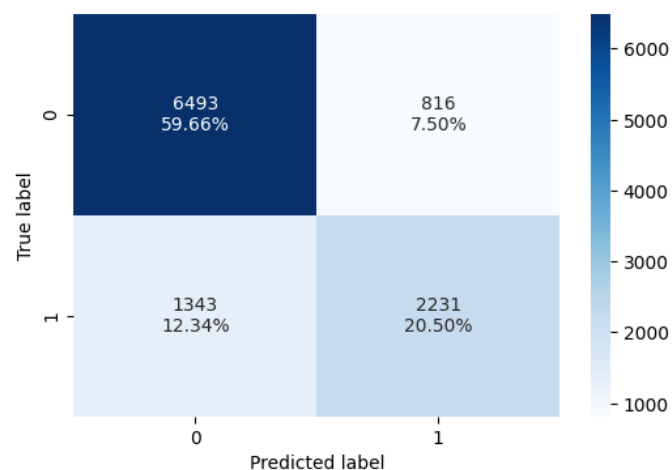
Checking Logistic Regression model performance on training set:

	Accuracy	Recall	Precision	F1
0	0.804387	0.627	0.736225	0.677237



Checking Logistic Regression model performance on test set

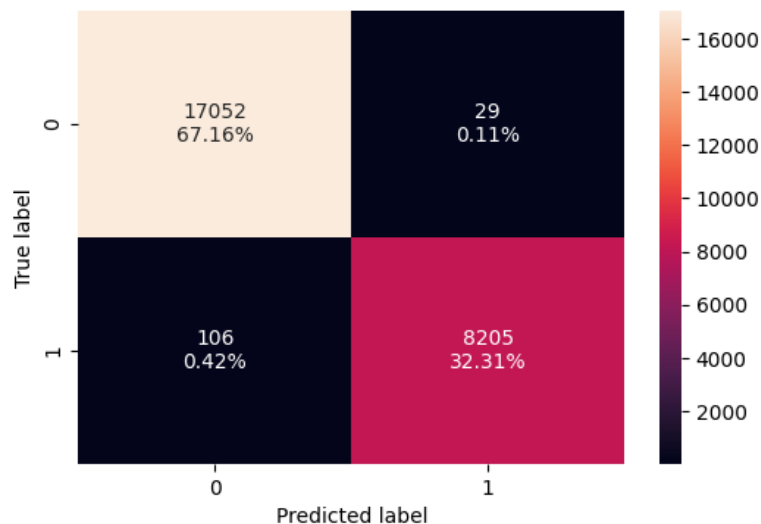
	Accuracy	Recall	Precision	F1
0	0.801617	0.624231	0.732196	0.673916



The Logistic Regression model has a average performance on both training and test sets

Decision Tree Classifier:

	Accuracy	Recall	Precision	F1
0	0.994683	0.987246	0.996478	0.99184



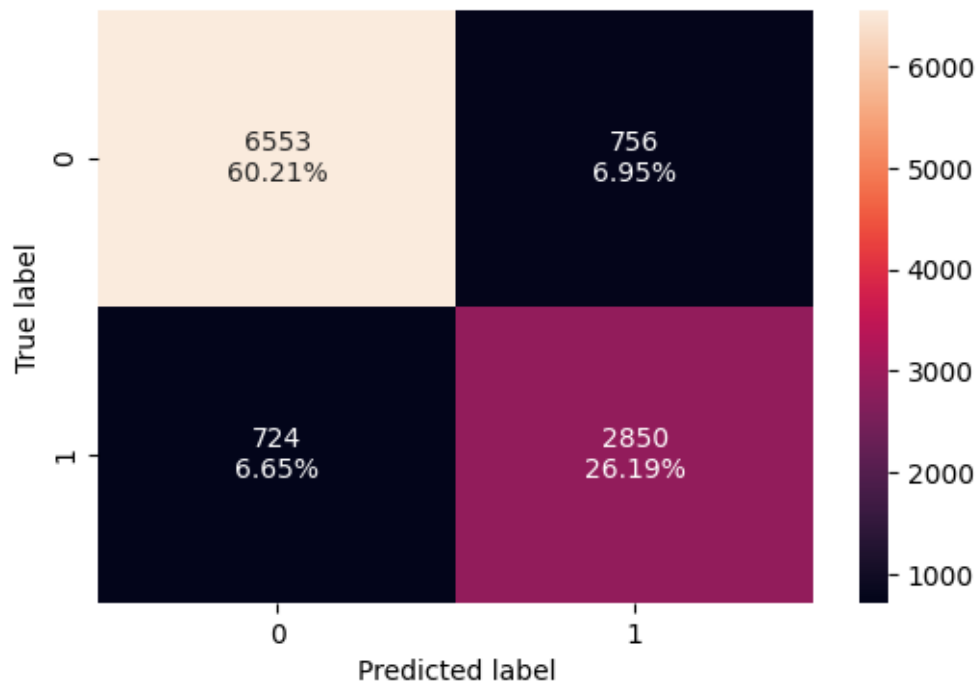
- Almost 0 errors on the training set, each sample has been classified correctly.
- Model has performed very well on the training set.
- As we know a decision tree will continue to grow and classify each data point correctly if no restrictions are applied as the trees will learn all the patterns in the training set.
- Let's check the performance on test data to see if the model is overfitting

Checking Decision Tree Classifier performance on test set

	Accuracy	Recall	Precision	F1
0	0.864008	0.797426	0.790349	0.793872

Observations

- The f1_score of the model is ~0.79 and we will try to maximize it further
- The variables used to build the model might contain multicollinearity, which will affect the p-values
- We will have to remove multicollinearity from the data to get reliable coefficients and p-values



Model Performance Improvement

Logistic Regression (deal with multicollinearity, remove high p-value variables, determine optimal threshold using ROC curve)

Model evaluation criterion:

Model can make wrong predictions as:

Predicting a booking cancellations as not cancelled and not cancelled as cancellations

Which case is more important?

- If we predict a guest does cancel, but the model predicts they will not. we might hold a room unnecessarily, turn away other potential guests, lose revenue due to last-minute vacancy. We may also lost opportunity and unpreparedness. This is called False negatives.
- If we predict a guest didnot cancel as canceled, we can offer the room to others, overbook to compensate. This is called False positive.

**predicting a cancellation as not canceled (false negative) is worse, because it can as it leads to unsold inventory, reduces revenue and disruptive planning. **

How to reduce this loss?

- We need to reduce False Negatives
- f1_score should be maximized as the greater the f1_score, the higher the chances of reducing False Negatives
 - fi_score is computed as

$$f1_score = \frac{2 * Precision * Recall}{Precision + Recall}$$

Detecting and Dealing with Multicollinearity:

There are different ways of detecting (or testing for) multicollinearity. One such way is using the Variation Inflation Factor (VIF).

- **Variance Inflation factor:** Variance inflation factors measure the inflation in the variances of the regression coefficients estimates due to collinearities that exist among the predictors. It is a measure of how much the variance of the estimated regression coefficient β_k is "inflated" by the existence of correlation among the predictor variables in the model.
- **General Rule of thumb:**
 - If VIF is 1 then there is no correlation among the kth predictor and the remaining predictor variables, and hence the variance of β_k is not inflated at all
 - If VIF exceeds 5, we say there is moderate multicollinearity
 - If VIF is equal or exceeding 10, it shows signs of high multi-collinearity

Inferences:

- market_segment_type_Online shows high multicollinearity too with VIF 69.9 and market_segment_type_Offline shows VIF 63.3.
- market_segment_type_Corporate shows moderate VIF as 16.74
- Dropping market_segment_type_Online due to high VIF

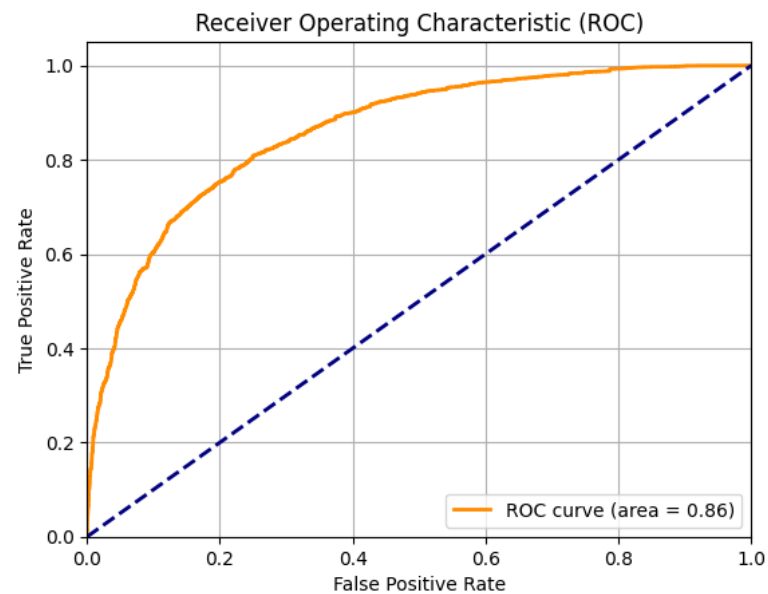
Removing high p-value variables:

- we will drop variables with high p-value one by one iteratively by using a loop
- This loop will repeat the steps for all the variables with p-value > 0.05

Final selected features:

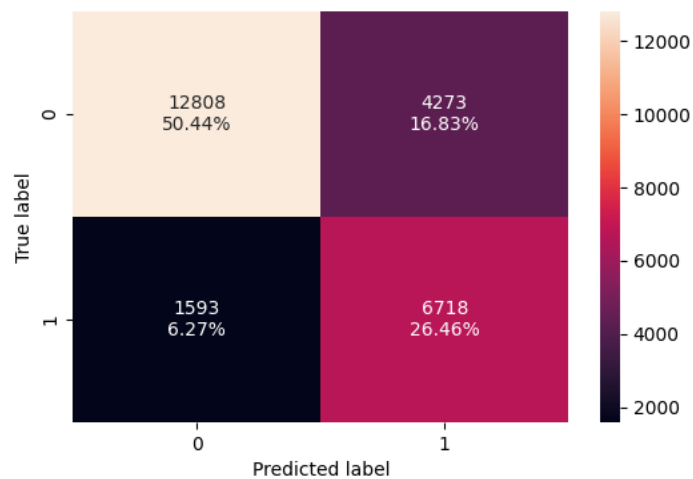
'const', 'no_of_adults', 'no_of_children', 'no_of_weekend_nights', 'no_of_week_nights', 'lead_time', 'arrival_year', 'arrival_month', 'no_of_previous_bookings_not_canceled', 'avg_price_per_room', 'no_of_special_requests', 'type_of_meal_plan_Not Selected', 'room_type_reserved_Room_Type 2', 'room_type_reserved_Room_Type 4', 'room_type_reserved_Room_Type 5', 'room_type_reserved_Room_Type 6', 'room_type_reserved_Room_Type 7', 'market_segment_type_Corporate', 'market_segment_type_Offline'

- We Training the Logistic Regression model again with only the significant features
- Determining optimal threshold using ROC Curve



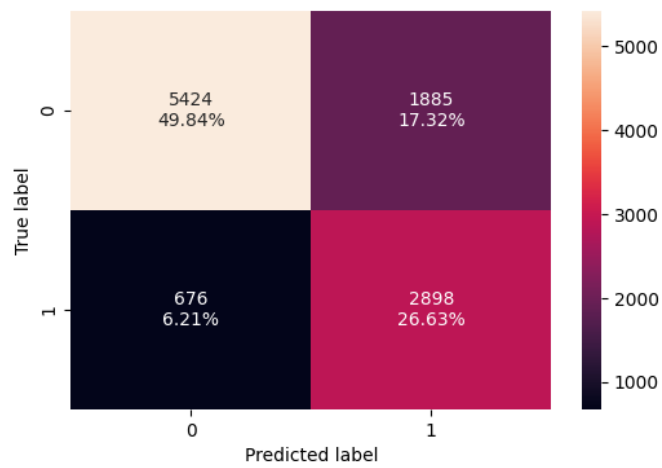
- Checking tuned Logistic Regression model performance on training set

	Accuracy	Recall	Precision	F1
0	0.768982	0.808326	0.611227	0.696094



- Checking tuned Logistic Regression model performance on test set

	Accuracy	Recall	Precision	F1
0	0.764679	0.810856	0.605896	0.69355



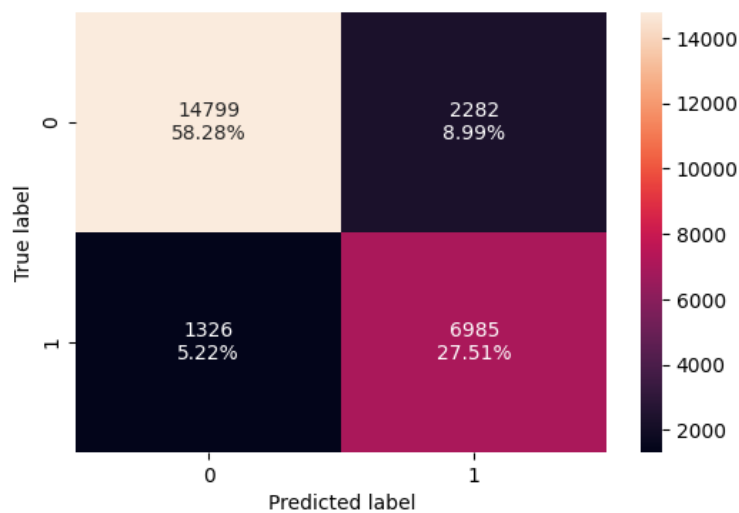
- The performance of the Logistic Regression model not changed (on both training and test sets) by dropping variables with high p-values and setting an optimal threshold.

Decision Tree Classifier (pre-pruning and post-pruning):

Pre - pruning the tree:

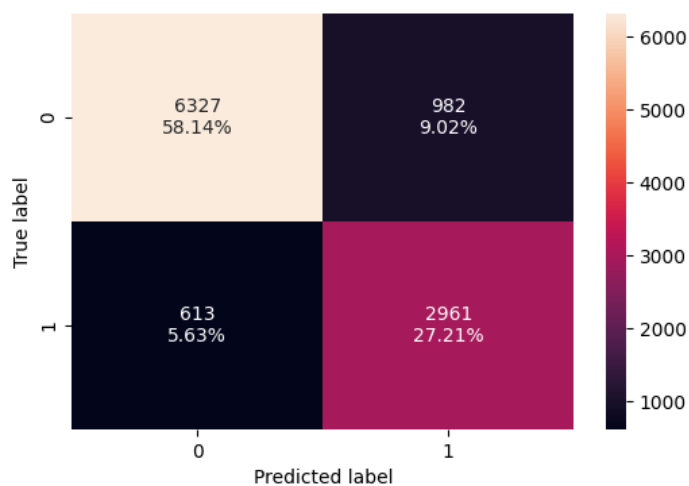
- Checking pre-pruned Decision Tree Classifier performance on training set

	Accuracy	Recall	Precision	F1
0	0.857908	0.840452	0.75375	0.794743

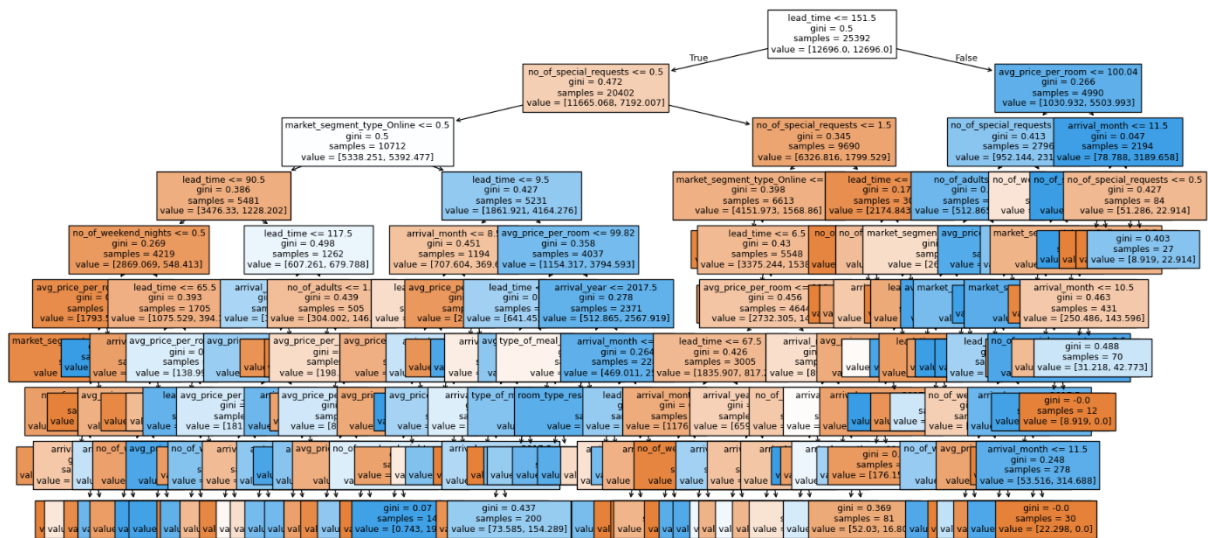


- Checking pre-pruned Decision Tree Classifier performance on test set

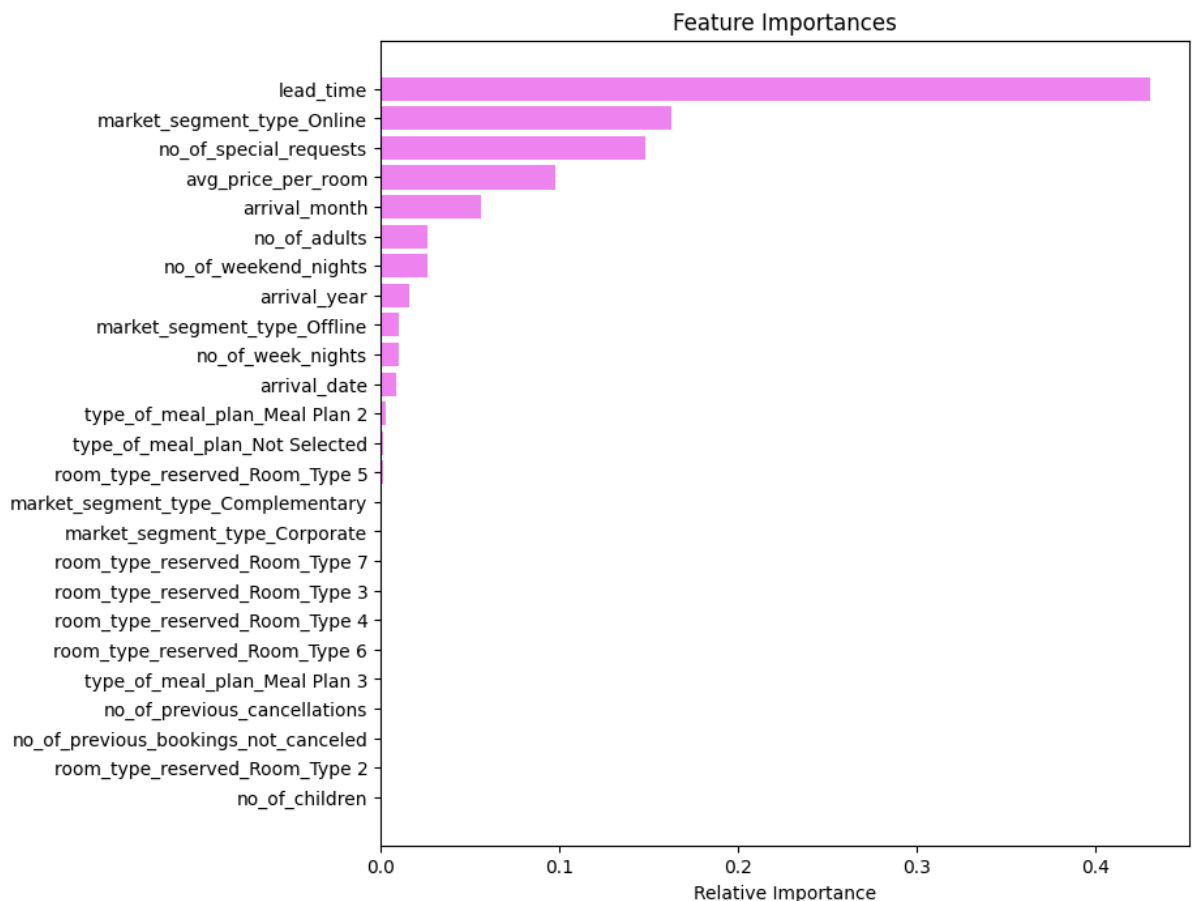
	Accuracy	Recall	Precision	F1
0	0.853441	0.828483	0.750951	0.787814



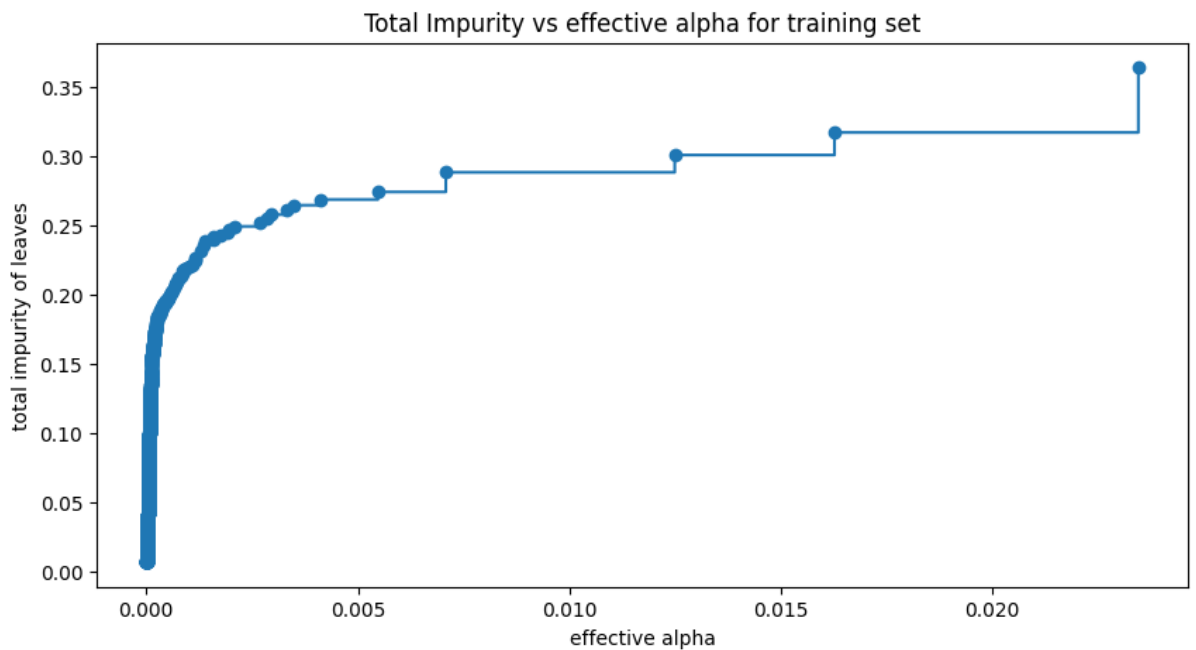
Visualizing the Decision Tree:



- Analyzing Feature Importance for tuned Decision Tree Classifier

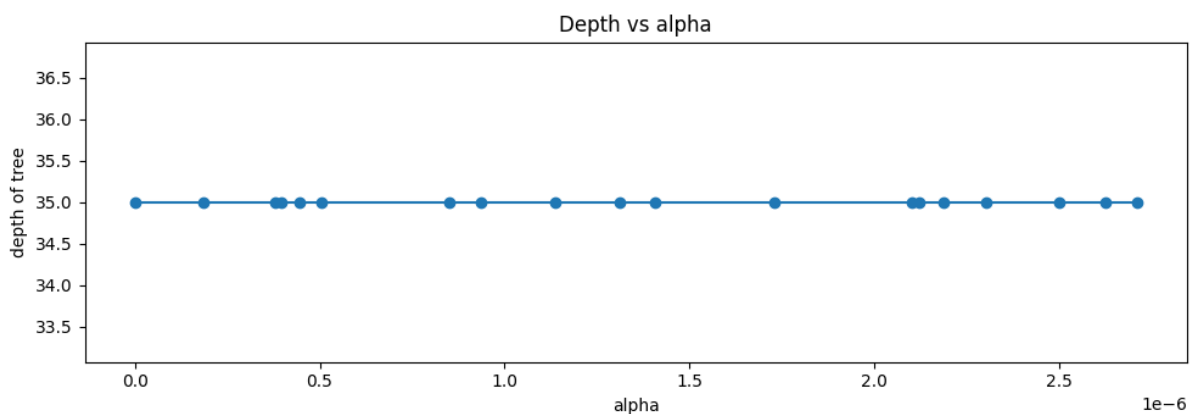
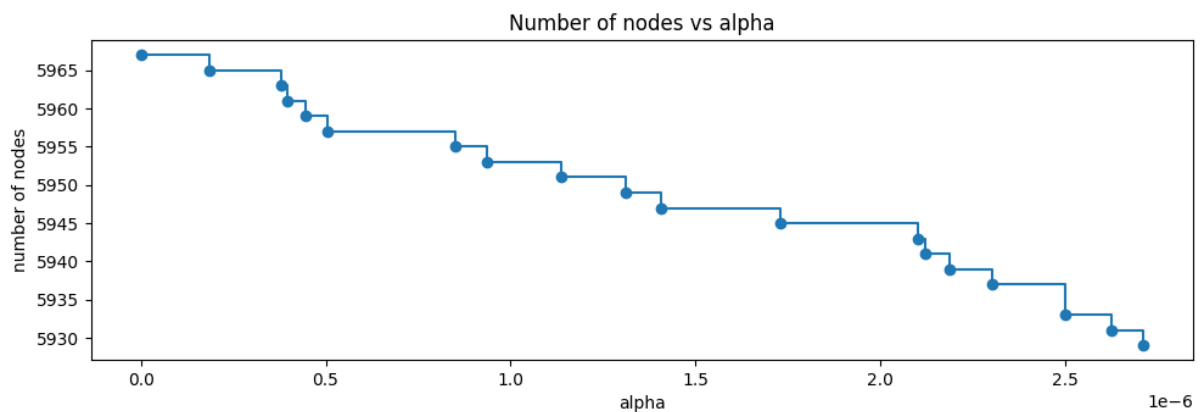


- Post - pruning the tree

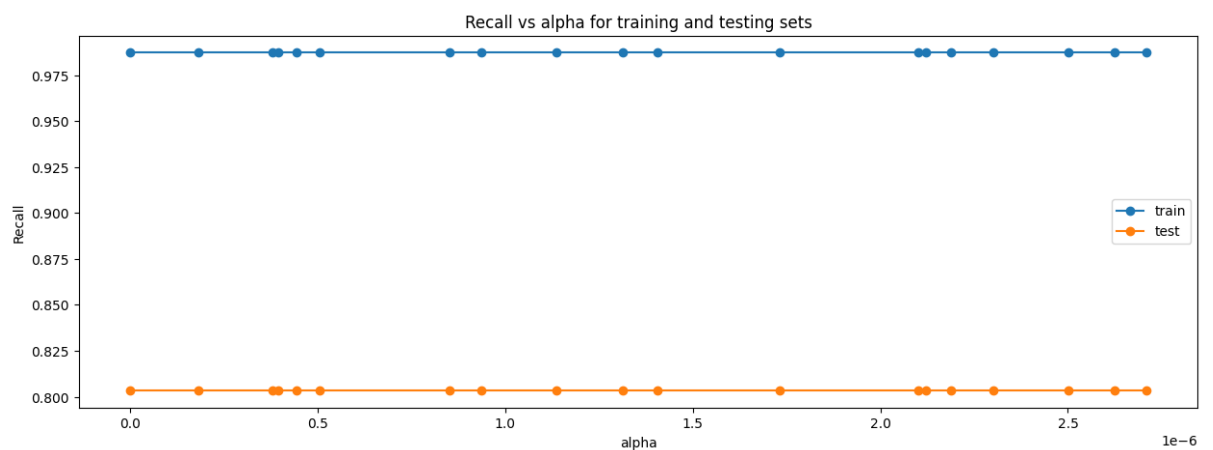


Next, we train a decision tree using effective alphas. The last value in `ccp_alphas` is the alpha value that prunes the whole tree, leaving the tree, `clfs[-1]`, with one node.

For the remainder, we remove the last element in `clfs` and `ccp_alphas`, because it is the trivial tree with only one node. Here we show that the number of nodes and tree depth decreases as alpha increases.

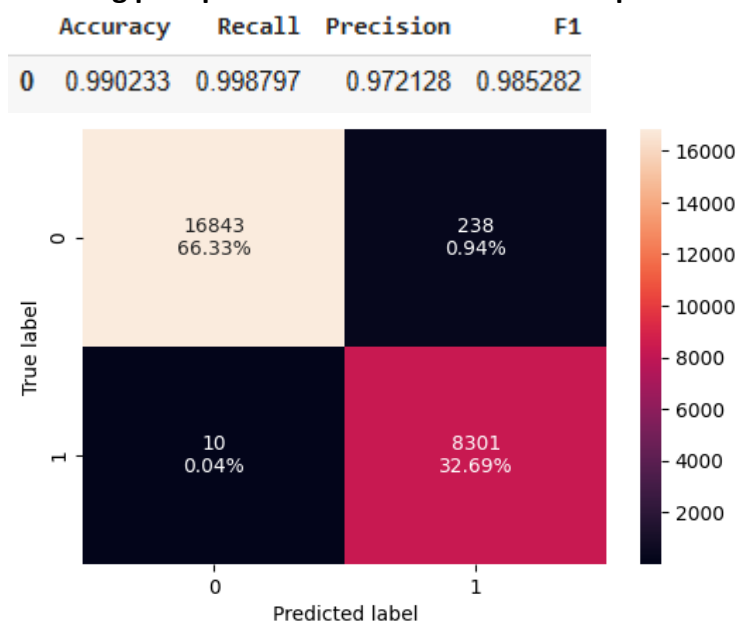


- Recall vs alpha for training and test sets



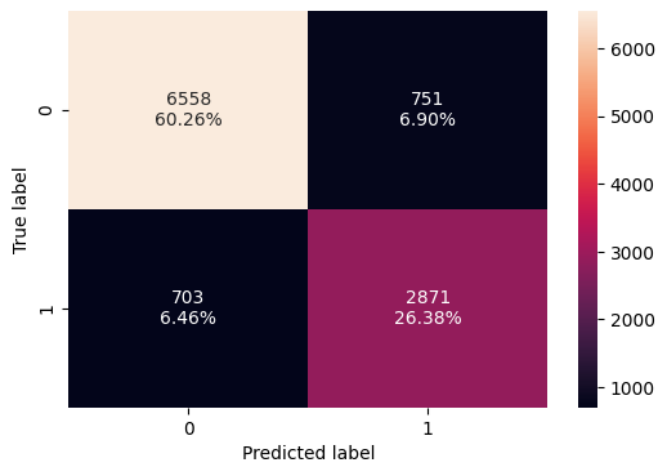
- No matter how much pruning you apply (ccp_alpha), recall on the training set remains extremely high, but the testing set recall does not improve

- Checking post-pruned Decision Tree Classifier performance on training set

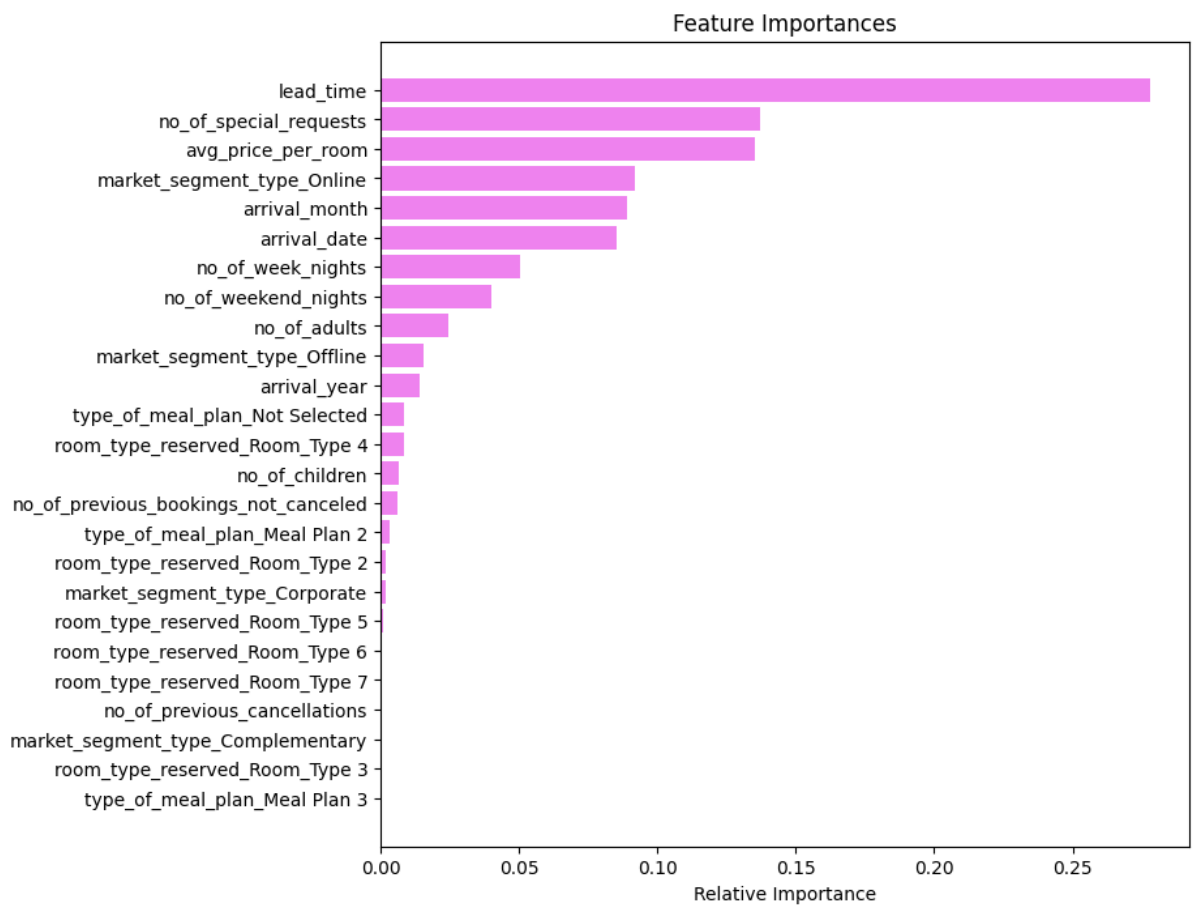


- Checking post-pruned Decision Tree Classifier performance on test set

	Accuracy	Recall	Precision	F1
0	0.865846	0.801903	0.79215	0.796997



Feature importances:



Model Performance Comparison and Final Model Selection:

Train set comparison:

Training performance comparison:

	Logistic Regression Base	Logistic Regression Improved	Decision Tree Base	Decision Tree Pre-Pruned	Decision Tree Post-Pruned
Accuracy	0.804387	0.768982	0.994683	0.857908	0.990233
Recall	0.627000	0.808326	0.987246	0.840452	0.998797
Precision	0.736225	0.611227	0.996478	0.753750	0.972128
F1	0.677237	0.696094	0.991840	0.794743	0.985282

Test set comparison:

Test set performance comparison:

	Logistic Regression Base	Logistic Regression Tuned	Decision Tree Base	Decision Tree Pre-Pruned	Decision Tree Post-Pruned
Accuracy	0.801617	0.764679	0.864008	0.853441	0.865846
Recall	0.624231	0.810856	0.797426	0.828483	0.801903
Precision	0.732196	0.605896	0.790349	0.750951	0.792150
F1	0.673916	0.693550	0.793872	0.787814	0.796997

Observations

- Decision tree model with default parameters is less overfitting the training data and is able to generalize well.
- Pre-pruned tree has given a generalized performance with the recall score of 0.84 and 0.82 on training and test set, respectively.

Actionable Insights and Recommendations

1. logistic regression model and odds ratio analysis reveal that:
 - Higher cancellation likelihood is associated with:
 - Longer lead times
 - Higher average room prices
 - More children
 - Meal Plan 2
 - Lower cancellation likelihood is strongly tied to:
 - Special requests (most significant protective factor)
 - Repeat bookings
 - Offline and corporate channels
 - Certain room types (Room Type 2, 5, and 6)
2. Threshold Tuning We evaluated thresholds between 0.25 and 0.5:
 - At threshold = 0.25, we achieved maximum recall (~84%), which aligns best with the business objective of minimizing false negatives.
 - Despite slightly lower precision, this threshold helps catch more actual cancellations.

Recommendation: Set deployment threshold at 0.25 and inform guests for early confirmations (like confirmation emails or flexible rescheduling offers).

3. Model Comparison Among logistic regression and decision tree variants:
 - Best balanced model: Post-Pruned Decision Tree as it shows high F1-score, precision, and accuracy
 - Highest recall: Pre-Pruned Decision Tree (82.8%) and Tuned Logistic (81.1%)Hence,
 - Using the Post-Pruned Tree as our main model
 - Using Logistic Regression (Tuned) as a secondary model.
4. Recall vs. Tree Complexity From the ccp_alpha plots:
 - Recall on the training set stays high, indicating overfitting
 - Test set recall doesnot improve with pruning thus suggesting basic pruning isnot enough
5. Recommendations

- Customer Engagement:
 - Personalize outreach for high-risk segments (long lead time, Meal Plan 2, families with children)
 - Offer incentives (e.g. discounts, flexibility) to guests showing cancellation but high lifetime value
- Operational Policy:
 - Allow more flexible policies or offer confirmations for repeat guests and corporates

-----END-----